GEBZE TEKNİK ÜNİVERSİTESİ

Lojik Devreler ve Tasarım

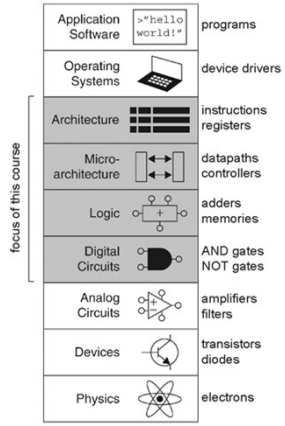
Bahar 2023

2. Sayı Sistemleri

1

Abstraction

Hiding details when they aren't important





2

Digital Discipline: Binary Values

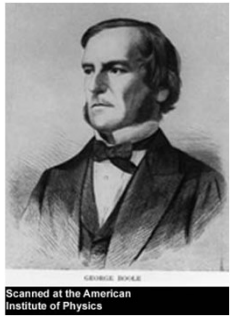
- **Two discrete values:**
  - 1's and 0's
  - 1, TRUE, HIGH
  - 0, FALSE, LOW
- Digital circuits use **voltage** levels to represent 1 and 0
- ***Bit*****:** *B*inary dig*it* - Smallest unit of information



3

George Boole, 1815-1864

- Born to working class parents
- Taught himself mathematics and joined the faculty of Queen's College in Ireland
- Wrote *An Investigation of the Laws of Thought* (1854)
- Introduced binary variables
- Introduced the three fundamental logic operations: AND, OR, and NOT





4

Number Systems

- Decimal numbers

How do we get the number representation of **5374** from the symbols 5,3,7, and 4?



Number Systems

- Decimal numbers

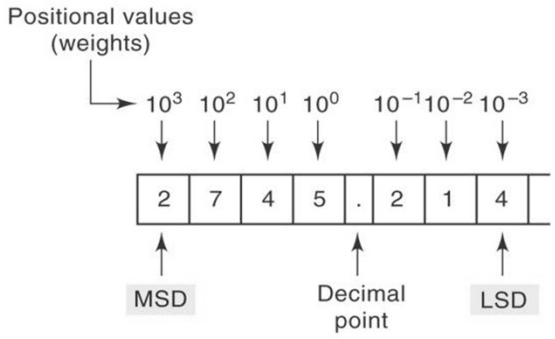
1's column
10's column
100's column
1000's column

$$5374_{10} = 5 \times 10^3 + 3 \times 10^2 + 7 \times 10^1 + 4 \times 10^0$$

five thousands; three hundreds; seven tens; four ones



Number Systems





Number Systems

- Binary numbers

How do we get the number representation of **1101** from the symbols 1 and 0?



Number Systems

- Binary numbers

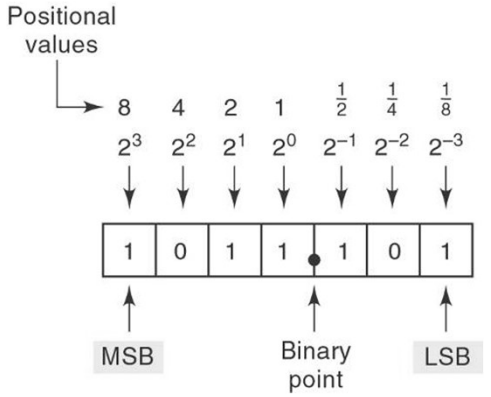
8's column 4's column 2's column 1's column

$1101_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 13_{10}$

one eight, one four, no two, one one



Number Systems



Powers of Two

Handy to memorize up to 2^9

n	2^n	n	2^n	n	2^n
0	1	8	256	16	65,536
1	2	9	512	17	131,072
2	4	10	1,024 (1K)	18	262,144
3	8	11	2,048	19	524,288
4	16	12	4,096 (4K)	20	1,048,576 (1M)
5	32	13	8,192	21	2,097,152
6	64	14	16,384	22	4,194,304
7	128	15	32,768	23	8,388,608



Number Conversion

- Binary to decimal conversion:
  - Convert 10011_2 to decimal
- Decimal to binary conversion:
  - Convert 47_{10} to binary



Number Conversion

- Binary to decimal conversion:
  - Convert 10011_2 to decimal
  - $16\times1 + 8\times0 + 4\times0 + 2\times1 + 1\times1 = 19_{10}$
- Decimal to binary conversion:
  - Convert 47_{10} to binary
  - $32\times1 + 16\times0 + 8\times1 + 4\times1 + 2\times1 + 1\times1 = 101111_2$



Decimal to Binary Conversion

- Two methods:
  - **Method 1:** Find the largest power of 2 that fits, subtract and repeat
  - **Method 2:** Repeatedly divide by 2, remainder goes in next most significant bit



Decimal to Binary Conversion

Method 1: Find the largest power of 2 that fits, subtract and repeat

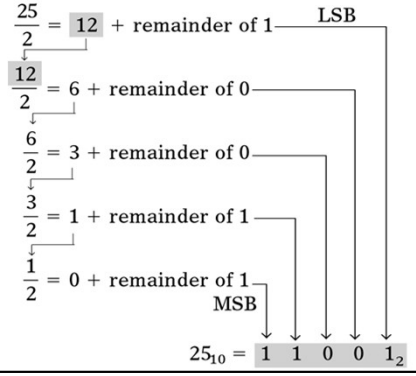
$$45_{10} = 32 + 8 + 4 + 1 = 2^5 + 0 + 2^3 + 2^2 + 0 + 2^0$$
$$= 1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1_2$$

$$76_{10} = 64 + 8 + 4 = 2^6 + 0 + 0 + 2^3 + 2^2 + 0 + 0$$
$$= 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0_2$$



Decimal to Binary Conversion

Method 2: Repeatedly divide by 2, remainder goes in next most significant bit





Decimal to Binary Conversion

$$\frac{37}{2} = 18.5 \longrightarrow \text{remainder of 1 (LSB)}$$

$$\frac{18}{2} = 9.0 \longrightarrow 0$$

$$\frac{9}{2} = 4.5 \longrightarrow 1$$

$$\frac{4}{2} = 2.0 \longrightarrow 0$$

$$\frac{2}{2} = 1.0 \longrightarrow 0$$

$$\frac{1}{2} = 0.5 \longrightarrow 1 \text{ (MSB)}$$



Decimal to Binary Conversion

Another example: Convert 75_{10} to binary.



Decimal to Binary Conversion

Another example: Convert 75_{10} to binary.

$75_{10} = 64 + 8 + 2 + 1 = 1001011_2$



Decimal to Binary Conversion

Another example: Convert 75_{10} to binary.

$75_{10} = 64 + 8 + 2 + 1 = 1001011_2$

or

75/2 = 37 remainder of 1
37/2 = 18 remainder of 1
18/2 = 9 remainder of 0
9/2 = 4 remainder of 1
4/2 = 2 remainder of 0
2/2 = 1 remainder of 0
1/2 = 0 remainder of 1



Binary Values and Range

- ***N*-digit decimal number**
  - How many values?
  - Range?
  - Example: 3-digit decimal number:
- ***N*-bit binary number**
  - How many values?
  - Range:
  - Example: 3-digit binary number:



Binary Values and Range

- ***N*-digit decimal number**
  - How many values? $\mathbf{10^N}$
  - Range? **[0, 10^N - 1]**
  - Example: 3-digit decimal number:
    - **10^3 = 1000 possible values**
    - **Range: [0, 999]**
- ***N*-bit binary number**
  - How many values? $\mathbf{2^N}$
  - Range: **[0, 2^N - 1]**
  - Example: 3-digit binary number:
    - **2^3 = 8 possible values**
    - **Range: [0, 7] = [000_2 to 111_2]**



Numbers with Different Bases

Decimal (base 10)	Binary (base 2)	Octal (base 8)	Hexadecimal (base 16)
00	0000	00	0
01	0001	01	1
02	0010	02	2
03	0011	03	3
04	0100	04	4
05	0101	05	5
06	0110	06	6
07	0111	07	7
08	1000	10	8
09	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F



Hexadecimal to Decimal

16^4	16^3	16^2	16^1	16^0	16^{-1}	16^{-2}	16^{-3}	16^{-4}

Hexadecimal point

$$356_{16} = 3 \times 16^2 + 5 \times 16^1 + 6 \times 16^0$$
$$= 768 + 80 + 6$$
$$= 854_{10}$$

$$2AF_{16} = 2 \times 16^2 + 10 \times 16^1 + 15 \times 16^0$$
$$= 512 + 160 + 15$$
$$= 687_{10}$$



Decimal to Hexadecimal

$$\frac{423}{16} = 26 + \text{remainder of } 7$$

$$\frac{26}{16} = 1 + \text{remainder of } 10$$

$$\frac{1}{16} = 0 + \text{remainder of } 1$$

$$423_{10} = 1A7_{16}$$

Hexadecimal to Binary Conversion

- Represent each symbol with 4-bit numbers
- Leading zeros can be added to the left to fill the last group

$$9F2_{16} = \begin{matrix} 9 & & F & & 2 \\ \downarrow & & \downarrow & & \downarrow \end{matrix}$$

$$= 1\ 0\ 0\ 1 \quad 1\ 1\ 1\ 1 \quad 0\ 0\ 1\ 0$$

$$= 100111110010_2$$

Hexadecimal to Binary Conversion

- Hexadecimal to binary conversion:
  - Convert $4AF_{16}$ (also written 0x4AF) to binary

Hexadecimal to Binary Conversion

- Hexadecimal to binary conversion:
  - Convert $4AF_{16}$ (also written 0x4AF) to binary
  - $0100\ 1010\ 1111_2$

Bits, Bytes, Nibbles…

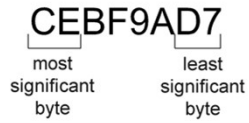
- Bits



- Bytes & Nibbles



- Bytes



IBM System/360 ('65 - '78)



Large Powers of Two

- 2^{10} = 1 kilo ≈ 1000 (1024)
- 2^{20} = 1 mega ≈ 1 million (1,048,576)
- 2^{30} = 1 giga ≈ 1 billion (1,073,741,824)

Estimating Powers of Two

- What is the value of 2^{24}?
- How many values can a 32-bit variable represent?

Estimating Powers of Two

- What is the value of 2^{24}?
  $2^4 \times 2^{20} \approx$ 16 million
- How many values can a 32-bit variable represent?
  $2^2 \times 2^{30} \approx$ 4 billion



Addition

- Decimal

```
  3734
+ 5168
```

- Binary

```
  1011
+ 0011
```



Addition

- Decimal

```
  11  ← carries
  3734
+ 5168
  8902
```

- Binary

```
  11  ← carries
  1011
+ 0011
  1110
```



Binary Addition Examples

- Add the following 4-bit binary numbers

```
  1001
+ 0101
```

- Add the following 4-bit binary numbers

```
  1011
+ 0110
```



Binary Addition Examples

- Add the following 4-bit binary numbers

```
   1
 1001
+0101
-----
 1110
```

- Add the following 4-bit binary numbers

```
  111
 1011
+0110
-----
10001
```


Binary Addition Examples

- Add the following 4-bit binary numbers

```
   1
 1001
+0101
-----
 1110
```

- Add the following 4-bit binary numbers

```
  111
 1011
+0110
-----
10001
```

Overflow!

Question

- How to represent Negative Numbers?

Signed Binary Numbers

- Sign/Magnitude Numbers
- Two's Complement Numbers

Sign/Magnitude Numbers

- 1 sign bit, *N*-1 magnitude bits
- Sign bit is the most significant (left-most) bit
  - Positive number: sign bit = 0
  - Negative number: sign bit = 1
- Example, 4-bit sign/mag representations of ± 6:
  +6 =
  - 6 =
- Range of an *N*-bit sign/magnitude number:



Sign/Magnitude Numbers

- 1 sign bit, *N*-1 magnitude bits
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- Example, 4-bit sign/mag representations of ± 6:
  +6 = **0110**
  - 6 = **1110**
- Range of an *N*-bit sign/magnitude number:
  $[-(2^{N-1}-1), 2^{N-1}-1]$



Sign/Magnitude Numbers

- 1 sign bit, *N*-1 magnitude bits
- Sign bit is the most significant (left-most) bit
  - Positive number: sign bit = 0
  - Negative number: sign bit = 1

$A: \{a_{N-1}, a_{N-2}, \ldots a_2, a_1, a_0\}$

$$A = (-1)^{a_{N-1}} \sum_{i=0}^{N-2} a_i \, 2^i$$

- Example, 4-bit sign/mag representations of ± 6:
  +6 = **0110**
  - 6 = **1110**
- Range of an *N*-bit sign/magnitude number:
  $[-(2^{N-1}-1), 2^{N-1}-1]$



Sign/Magnitude Numbers

Problems:

- Addition doesn't work, for example -6 + 6:

  1110
  + 0110
  10100 (wrong!)

- Two representations of 0 (± 0):

  1000
  0000



Two's Complement Numbers

- Don't have same problems as sign/magnitude numbers:
  - **Addition works**
  - **Single representation for 0**



Two's Complement Numbers

- msb has value of -2^{N-1}

$$A = a_{N-1}(-2^{N-1}) + \sum_{i=0}^{N-2} a_i\, 2^i$$

- Most positive 4-bit number:
- Most negative 4-bit number:
- The most significant bit still indicates the sign (1 = negative, 0 = positive)
- Range of an *N*-bit two's complement number:



Two's Complement Numbers

- msb has value of -2^{N-1}

$$A = a_{N-1}(-2^{N-1}) + \sum_{i=0}^{N-2} a_i\, 2^i$$

- Most positive 4-bit number: **0111**
- Most negative 4-bit number: **1000**
- The most significant bit still indicates the sign (1 = negative, 0 = positive)
- Range of an *N*-bit two's complement number:

$[-(2^{N-1}), 2^{N-1}-1]$



"Taking the Two's Complement"

- "Taking the Two's complement" **flips the sign** of a two's complement number
- **Method:**
  1. Invert the bits
  2. Add 1
- **Example:** Flip the sign of $3_{10} = 0011_2$



"Taking the Two's Complement"

- "Taking the Two's complement" **flips the sign** of a two's complement number
- **Method:**
  1. Invert the bits
  2. Add 1
- **Example:** Flip the sign of $3_{10} = 0011_2$
  1. **1100**
  2. **+ 1**

  $1101 = -3_{10}$



Two's Complement Examples

- Take the two's complement of $6_{10} = 0110_2$
- What is the decimal value of the two's complement number 1001_2?



Two's Complement Examples

- Take the two's complement of $6_{10} = 0110_2$
  1. 1001
  2. + 1

  $1010_2 = -6_{10}$
- What is the decimal value of the two's complement number 1001_2?
  1. 0110
  2. + 1

  $0111_2 = 7_{10}$, so $1001_2 = -7_{10}$



Two's Complement Addition

- Add 6 + (-6) using two's complement numbers

$$\begin{array}{r} 0110 \\ +\ 1010 \\ \hline \end{array}$$

- Add -2 + 3 using two's complement numbers

$$\begin{array}{r} 1110 \\ +\ 0011 \\ \hline \end{array}$$



Two's Complement Addition

- Add 6 + (-6) using two's complement numbers

```
  111
   0110
 + 1010
 -------
  10000
```

- Add -2 + 3 using two's complement numbers

```
  111
   1110
 + 0011
 -------
  10001
```



Increasing Bit Width

Extend number from *N* to *M* bits (*M* > *N*) :

- Sign-extension
- Zero-extension



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Sign-Extension

- Sign bit copied to msb's
- Number value is same
- **Example 1:**
  - 4-bit representation of 3 = **0**011
  - 8-bit sign-extended value: **0000**0011
- **Example 2:**
  - 4-bit representation of -5 = **1**011
  - 8-bit sign-extended value: **1111**1011



Zero-Extension

- Zeros copied to msb's
- Value changes for negative numbers
- **Example 1:**
  - 4-bit value = $0011 = 3_{10}$
  - 8-bit zero-extended value: **0000**0011 = 3_{10}
- **Example 2:**
  - 4-bit value = $1011 = -5_{10}$
  - 8-bit zero-extended value: **0000**1011 = 11_{10}



Number System Comparison

Decimal	Signed-2's Complement	Signed-1's Complement	Signed Magnitude
+7	0111	0111	0111
+6	0110	0110	0110
+5	0101	0101	0101
+4	0100	0100	0100
+3	0011	0011	0011
+2	0010	0010	0010
+1	0001	0001	0001
+0	0000	0000	0000
−0	—	1111	1000
−1	1111	1110	1001
−2	1110	1101	1010
−3	1101	1100	1011
−4	1100	1011	1100
−5	1011	1010	1101
−6	1010	1001	1110
−7	1001	1000	1111
−8	1000	—	—



Binary Codes – Binary Coded Decimal

- A binary code is a representation of a given set.
- An example is Binary Coded Decimal (BCD).
- Each group of 4-bits represents one decimal digit.
- Binary combinations 1010 through 1111 are not used and have no meaning in BCD



Binary Codes – Binary Coded Decimal

8	7	4	(decimal)
↓	↓	↓	
1000	0111	0100	(BCD)

9	4	3	(decimal)
↓	↓	↓	
1001	0100	0011	(BCD)

Decimal Symbol	BCD Digit
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001



Binary Codes – Binary Coded Decimal

Convert **0110100000111001** (BCD) to its decimal equivalent.

$$\underbrace{0110}_{6}\ \underbrace{1000}_{8}\ \underbrace{0011}_{3}\ \underbrace{1001}_{9}$$

Divide the BCD number into four-bit groups and convert each to decimal.



Binary Codes – Gray Code

- Over the straight binary number sequence, only one bit changes in each step.
- Used in applications where the normal sequence of binary numbers by the hardware may produce an error during transition between numbers.

Gray Code	Decimal Equivalent
0000	0
0001	1
0011	2
0010	3
0110	4
0111	5
0101	6
0100	7
1100	8
1101	9
1111	10
1110	11
1010	12
1011	13
1001	14
1000	15



Binary Codes

Decimal	Binary	Hexadecimal	BCD	GRAY
0	0	0	0000	0000
1	1	1	0001	0001
2	10	2	0010	0011
3	11	3	0011	0010
4	100	4	0100	0110
5	101	5	0101	0111
6	110	6	0110	0101
7	111	7	0111	0100
8	1000	8	1000	1100
9	1001	9	1001	1101
10	1010	A	0001 0000	1111
11	1011	B	0001 0001	1110
12	1100	C	0001 0010	1010
13	1101	D	0001 0011	1011
14	1110	E	0001 0100	1001
15	1111	F	0001 0101	1000



Binary Codes American Standard Code for Information Interchange (ASCII)

Character	HEX	Decimal	Character	HEX	Decimal	Character	HEX	Decimal	Character	HEX	Decimal
NUL (null)	0	0	Space	20	32	@	40	64	.	60	96
Start Heading	1	1	!	21	33	A	41	65	a	61	97
Start Text	2	2	"	22	34	B	42	66	b	62	98
End Text	3	3	#	23	35	C	43	67	c	63	99
End Transmit.	4	4	$	24	36	D	44	68	d	64	100
Enquiry	5	5	%	25	37	E	45	69	e	65	101
Acknowlege	6	6	&	26	38	F	46	70	f	66	102
Bell	7	7	`	27	39	G	47	71	g	67	103
Backspace	8	8	(	28	40	H	48	72	h	68	104
Horiz. Tab	9	9	)	29	41	I	49	73	i	69	105
Line Feed	A	10	*	2A	42	J	4A	74	j	6A	106
Vert. Tab	B	11	+	2B	43	K	4B	75	k	6B	107
Form Feed	C	12	,	2C	44	L	4C	76	l	6C	108
Carriage Return	D	13	-	2D	45	M	4D	77	m	6D	109
Shift Out	E	14	.	2E	46	N	4E	78	n	6E	110

